
Software Requirements Specification

for

Project A.M.I.

Version 1.4.0 approved

**Prepared by: Stefano Buglione,
Lauryn Brown,
Marie Schwarz,
Nicolas Fernandez**

Team TheParents

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Revision History

Name	Date	Reason For Changes	Version
Marie Schwarz	02/18/26	Creation of SRS	1.0.0
Nicolas Fernandez	04/06/26	Updated SRS to expected format	1.1.0
Marie Schwarz	04/07/26	Updated content on sections: 1.1-1.4, 2.3-2.7, 3.1-3.2	1.1.5
Stefano Buglione	04/08/26	Updated content on sections: 5.2-5.5, Appendix A	1.2.0
Nicolas Fernandez	04/09/26	Fixed punctuation and grammatical issues	1.2.5
Marie Schwarz	04/10/26	Updated content on sections: 1.2, 3.3-3.4, 4	1.3.0
Laurn Brown	04/10/26	Update content and fixed punctuation/grammar on sections: 2.1-2.3, 2.7, 3.1, 4.1, 5.1, Appendix A	1.3.5
Nicolas Fernandez	04/10/26	Fixed formatting on all sections	1.4.0

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document is for the Artificial Medical Intelligence (**A.M.I.**) website and its corresponding AI model.

This SRS defines the software requirements for the first release of **A.M.I.** as a web-based prototype. It covers the public-facing website, the chatbot interaction system, source transparency features, citation display, user account capabilities for saving chats, and the supporting administrative/development considerations necessary for implementation and testing.

The following sections in this SRS cover **A.M.I.** version 1.0.0 (AMlv1.0.0).

1.2 Document Conventions

The default text font used in this SRS is Arial with font size 11. For a list, any relevant headers will be in font Arial Bold with its content in font Arial Italic. For a table, the header (1st row) will be highlighted in grey with font Arial Bold. A convention (term, requirement ID, or priority level) will be in font Arial Bold Italic.

Any important terminology used will be in font Arial Bold Italic.

The SRS uses the following conventions:

Term	Description
SHALL	Specifies a mandatory system requirement that must be implemented to allow the software product to function and follow relevant guidelines.
WILL	Refers to an expected system requirement for the software product to interact with users or systems.
MUST	A required system requirement for the software product to work within relevant restrictions within the given scope.
MAY	Can refer to: (1) An optional feature that can be implemented (2) An optional functionality that can be used by a user (3) Potential restriction that can affect performance
DEPENDS	A necessary system requirement for the software product to run.

The requirements are categorized as follows:

Format	Requirement
FRQ-XX	Functional
RS-XX	Response Sequences

The priorities are classified as follows:

Priority Level	Description
High	Critical requirements that are essential for the system to function. These must be

	implemented for the system to be usable and to meet its core objectives.
Medium	Important requirements that improve functionality and user experience. These are not required for the system to operate at a basic level. These can be implemented after high-priority features.
Low	Optional or enhancement features that provide additional value but are not necessary for initial system functionality. These may be implemented in future versions.

1.3 Intended Audience and Reading Suggestions

This document is intended for:

- *Developers, who will design and implement the system*
- *Testers, who will validate that requirements are met*
- *Instructors/evaluators, who will review the system as an academic project*
- *Documentation writers, who may prepare user-facing guides*
- *Users, who want to understand the product's purpose and scope*

Readers should begin with the Introduction and Overall Description sections for context. Developers and testers should focus next on External Interface Requirements, System Features, and Other Non-functional Requirements. Users should pay particular attention to Scope, Constraints, Assumptions, and Dependencies.

1.4 Product Scope

A.M.I. is a web-based AI assistant that can conduct text-based conversations on topics related to personalized medicine. It is designed to combat the issue against the “black box” nature of **LLMs** and AI models through source transparency.

A.M.I. WILL aid the public in understanding how AI models use sources to generate responses, reduce misinformation, increase accountability of AI responses, and provide educative insights on topics in personalized medicine.

In Scope:

- *A web-based chatbot interface for text-based interaction*
- *AI-generated responses related to personalized medicine and general health education*
- *Display of sources used to generate responses*
- *Citation formatting in **APA 7** style*
- *Standard disclosure message in all responses stating that **A.M.I.** is not a medical professional*

- *User registration and login for saving chat history*
- *Viewing previous saved chats for registered users*
- *Basic prompt validation and out-of-scope response handling*
- *Source-aware follow-up prompting*
- *Secure storage of account information and chat history*
- *Responsive web design for desktop and laptop use*
- *Administrative/developer tools for debugging, testing, and maintenance*

Out of Scope:

- *Diagnosis, treatment, or prescription of medical conditions*
- *Emergency medical advice or crisis response*
- *Integration with hospital systems, clinics, or insurance providers*
- *Access to private medical records or patient files*
- *Real-time monitoring of patient health data*
- *Upload and interpretation of lab reports, scans, or medical images*
- *Voice-based interaction in the first release unless later added as an enhancement*
- *Guaranteed medical accuracy equivalent to a licensed physician*
- *Topics unrelated to medicine, health education, or personalized medicine*

1.5 References

Contact Nicolas Fernandez through university email (ngf22@fsu.edu) to receive access to the GitHub repository to get copies of all the documents mentioned.

Documents:

Title	Author(s)	Date	Location
Functional Requirements	Stefano Buglione	02/16/26	GitHub – Phases – Requirements
Non-Functional Requirements	Lauryn Brown	02/16/26	GitHub – Phases – Requirements
User Roles	Nicolas Fernandez	02/19/26	GitHub – Phases – Requirements
AI Avatar Requirements	Stefano Buglione	02/16/26	GitHub – Phases – Requirements
Analysis of existing medical systems	Nicolas Fernandez	02/27/26	GitHub – Phases – Analysis
Bias and Hallucination Risk Analysis	Marie Schwarz	03/02/26	GitHub – Phases – Analysis
Dataset and Source Evaluation	Stefano Buglione	02/27/26	GitHub – Phases – Analysis
LLM Training Pipeline and Deployment Design	Marie Schwarz	03/07/26	GitHub – Phases – Design
System Architecture Design	Marie Schwarz	03/07/26	GitHub – Phases – Design
A.M.I. Chatbot Framework	Stefano Buglione	04/09/26	GitHub – Phases – Implementation
Test Plan	Team TheParents	04/09/26	GitHub – Phases – Test

Web Addresses:

Title	Author(s)	Date	Source
Text-To-Speech	VRCWizard	05/15/22	https://ttsvoicewizard.com/docs/intro
Software Requirement Specification (SRS) Format	Madhuri Hammad	09/20/23	https://www.geeksforgeeks.org/software-engineering/software-requirement-specification-srs-format/?limit=10
AI Accountability and Personal Data Protection Act	119 th Congress	07/21/25	https://www.congress.gov/bill/119th-congress/senate-bill/2367/text

2. Overall Description

2.1 Product Perspective

The **A.M.I.** software product is a new system created to address how **LLMs** and AI models lack transparency when responding to prompts. The software product operates as a standalone web-based system.

A.M.I. is designed using a client-server architecture, where a web-based frontend communicates with a backend server that handles user authentication, data storage, and AI response generation. The system integrates a trained **LLM** with a source retrieval pipeline to provide responses that include supporting references.

Product has the following components:

- **Front-end**
 - Built as a web-based user interface using **HTML, CSS, Python and JavaScript**
- **Backend:**
 - Handles **API** routes for chat generation, authentication, and chat management, also processes user prompts and responses
- **AI Model and Processing Layer**
 - Integrates retrieval mechanism (**PubMed, CDC/NIH**) to enhance transparency
 - Utilizes trained **LLM** (trained with **Unsloth**, base model is **LLaMa**-based)
- **Database System**
 - Stores user account information securely (**MongoDB**)
 - Stores chat histories and messages
 - Allows for retrieval of previous conversations
- **External Interface**
 - Public medical research databases (**PubMed** using **Entrez API**)
 - Web-hosted AI model (**HuggingFace**-compatible formats)

2.2 Product Functions

The **A.M.I.** system provides the following major functional capabilities:

- **User Interaction & Chat**
 - Accept user text input through a web-based chat interface

- *Generate AI responses related to personalized medicine and health education*
- *Maintain conversational context within a chat session*
- *Support follow-up questions with awareness of prior context*
- **Transparency & Source Integration**
 - *Retrieve relevant medical sources to support AI responses*
 - *Display citations alongside responses in **APA 7** format*
 - *Provide source-aware prompting and explanation of how information is derived*
- **Prompt Handling & Safety**
 - *Validate user input for relevance to in-scope topics*
 - *Detect and handle out-of-scope or unsafe queries*
 - *Provide standardized disclaimers indicating that **A.M.I.** is not a medical professional*
 - *Avoid diagnosis, prescriptions, or medical decision-making*
- **User Account Management**
 - *Allow users to register and log into the system*
 - *Securely store user credentials using encryption/hashing*
 - *Manage user sessions for authenticated access*
- **Chat History Management**
 - *Save chat conversations to a database*
 - *Allow users to view previous chats*
 - *Load and continue past conversations*
- **Avatar & UI Feedback**
 - *Display a **Live2D** avatar that reacts to system states*
 - *Provide visual feedback for:*
 - *Thinking/processing*
 - *Response generation*
 - *Urgent or serious responses*
 - *Update system status indicators (e.g., “Generating response...”)*
- **System & Developer Support**
 - *Provide backend endpoints for debugging and testing*
 - *Log errors and system behavior for maintenance*
 - *Support modular updates to AI model and retrieval components*

2.3 User Classes and Characteristics

The **A.M.I.** system **WILL** serve the following user classes based on role, domain knowledge, and expected usage context:

- 1) **General Public**
 - *Have a basic understanding of what AI is*
 - *Know how to interact with a text-based conversational AI model*
 - *Have an interest in medicine or general health*
 - *Requirements: User-friendly menu, clear navigation and readable responses, easy-to-understand disclaimer language, accessible citation display, guidance when prompts are out of scope*
- 2) **Students**
 - *Ages between 15-30 years, are in high school or college*
 - *Have a basic or advanced understanding of what AI is*
 - *Interest in or studies medicine*

- Know how to interact with a text-based conversational AI model
 - Have a basic understanding of how references are cited and how to access links or research the source
 - Requirements: Easy access to cited references, educational wording in responses, clear structure for follow-up questions, ability to save chats for later review
- 3) **Medical Professionals**
- Are practicing medicine
 - Have a basic or advanced understanding of what AI is
 - Know how to interact with a text-based conversational AI model
 - Understand how references are cited and how to formulate follow-up questions based on the cited source material
 - **MAY** use the system for quick informational support, not diagnosis
 - Can better interpret citation quality and context
 - Requirements: Transparent source display, concise but informative responses, easy ability to verify references, consistent disclaimer and boundary enforcement
- 4) **Researchers**
- Are in or adjacent to the medical or tech fields
 - Have an advanced understanding of what AI is
 - Know how to interact with a text-based conversational AI model
 - Understand how references are cited and how to formulate follow-up questions based on the cited source material
 - **MAY** be interested in the transparency and source-tracing aspects of the system
 - Likely to assess response quality critically
 - Requirements: Clear citation traceability, consistent source formatting, reliable response structure, better visibility into how information is grounded

2.4 Operating Environment

The system can operate in the following environments:

- 1) **Technical**
- A web-based application that is available via browsers
 - Responsive web design can be accessed on computers and laptops
 - Internet connection required for AI/**API** communication
 - Hosted in a development or cloud-based environment
- 2) **Hardware**
- User device with keyboard, screen, and internet connectivity
 - Server or cloud host capable of handling frontend, backend, and database services
 - No specialized medical hardware required
- 3) **Software**
- Supported browsers: Google Chrome, Safari, Microsoft Edge, Mozilla Firefox
 - **Flask** will be used for the backend framework for the website
 - **MongoDB** database with **NoSQL**
 - AI model integration through **API** or embedded model service
 - Operating systems: Windows, macOS, Linux

4) User

- *Is only able to be accessed via computer or laptops*
- *Can handle different operating systems and browser versions*
- *Requires internet access*

2.5 Design and Implementation Constraints

Design Constraints:

- *The system **MUST** remain within the educational and informational scope of personalized medicine*
- *The system **MUST** not claim to replace licensed medical professionals*
- *The project **MAY** be constrained by academic deadlines and prototype-level resources*

Implementation Constraints:

- *The quality of outputs depends on the selected **LLM/API***
- *Citation quality is dependent on source retrieval and formatting logic*
- *The initial release **MAY** prioritize desktop/laptop compatibility over full mobile optimization*
- *Security and privacy measures **MUST** be implemented for user accounts and saved chats*
- *The system **MUST** use understandable, non-misleading disclaimers*
- *The implementation **MAY** depend on specific frameworks, **APIs**, or hosting services chosen by the team*

2.6 User Documentation

This is what will be covered in the user manual that will be provided:

- *User guide for navigating the website*
- *Basic tutorial on using the chatbot*
- *Explanation of citations and source transparency*

2.7 Assumptions and Dependencies

Assumptions:

- *Users **MUST** have internet access and a modern browser*
- *Users understand that **A.M.I.** is for informational and educational purposes only*
- *Users can input prompts in readable text form*
- *The AI model used by the system **WILL** remain available during development and testing*
- *Users seeking real medical treatment are expected consult licensed professionals*
- *Users **WILL** be able to access system using desktop or laptop device*

Dependencies:

- *The system **DEPENDS** on the availability and functionality of the **MongoDB** database for storing user accounts and chat history*
- *The system **DEPENDS** on sufficient hardware resources (**GPU/VRAM** constraints and system memory) to support loading and running the AI model locally*
- *The system **DEPENDS** on the **Flask** backend framework to handle routing, **API** requests, and server-side processing*

- The system **DEPENDS** on the AI model (**Unsloth / LLaMA-based model**) being properly loaded and accessible for generating responses
- The system **DEPENDS** on external medical data sources (e.g., **PubMed** via **Entrez API**, **NIH/CDC** resources) for retrieving supporting references
- The system **DEPENDS** on third-party Python libraries (e.g., **Transformers**, **PEFT**, **PyTorch**, **Biopython**) for AI processing and data retrieval
- The system **DEPENDS** on client-side **JavaScript** libraries (e.g., **PixiJS** and **Live2D**) for rendering the avatar interface
- The system **DEPENDS** on the operating environment (**Python 3.12**, **Node/NPM** where applicable) being correctly configured
- The system **DEPENDS** on internet connectivity for accessing external **APIs** and loading required resources (e.g., **CDN**-hosted libraries)
- The system **DEPENDS** on browser compatibility (e.g., **Chrome**, **Microsoft Edge**, **Mozilla Firefox**) to properly render the **UI** and avatar components

3. External Interface Requirements

3.1 User Interfaces

The website **SHALL** provide interfaces for the following user types:

1) **Guest User**

- The website **WILL** allow unregistered (guest) users to utilize the **A.M.I.** AI model
- The website **WILL** allow users to register an account
- The website **SHALL** inform guest users the features that require registration

2) **Logged-In User**

- The website **SHALL** allow registered users to log into their accounts securely
- The website **SHALL** allow users to log out securely
- The website **SHALL** allow users to delete their account
- The website **SHALL** allow users to create new chat sessions
- The website **SHALL** save and retrieve recent chat history associated with the user account
- The website **SHALL** allow users to continue conversations from prior sessions
- The website **SHALL** allow users to update their preferences when interacting with the **A.M.I.** AI model in settings
- The website **SHALL** allow users to update to use text-to-speech to generate prompts

All user interfaces adhere to the following requirements:

- The website **WILL** have a user-friendly **UI/UX** interface to interact with the **A.M.I.** AI model
- The website **MUST** allow users to enter and exist guest mode
- The website **SHALL** allow users to read the about us page
- The website **SHALL** allow users to change the light/dark mode in settings
- The website **SHALL** display medical disclaimers in each chatbot response
- The website **SHALL** display AI responses with properly formatted citations

3.2 Hardware Interfaces

A.M.I. does not require specialized hardware interfaces to interact with the website nor with the AI model.

Supported device types:

- *Desktops*
- *Laptops*

Browser-based input/output components:

- *Keyboard*
- *Mouse/trackpad*
- *Screen display*
- *Internet-enabled network adapter*

Communication protocols used:

- **HTTP**
- **HTTPS**

3.3 Software Interfaces

The **A.M.I.** system interfaces with multiple software components that support its core functionality, including frontend, backend, AI processing, and data storage.

Frontend Interface:

- *Web-based user interface built using **HTML**, **CSS**, **Python**, and **JavaScript** frameworks*
- *Communicates with backend via **HTTP/HTTPS** requests*
- *Sends user prompts and receives AI-generated responses*

Backend Interface:

- *Application server through **Flask***
- *Handles request processing, authentication, and business logic*
- *Acts as an intermediary between frontend and AI model/**API***

AI Model / API Interface:

- *Integration with an **LLM API** for the **A.M.I.** avatar model*
- *Receives user prompts as input*
- *Returns generated responses as output*
- *May include additional metadata such as confidence scores or source references*

Database Interface:

- Database system (**MongoDB** with **NoSQL**)
- Stores:
 - User account data (username, password, authentication info)
 - Saved chat history
- Supports **CRUD** operations (Create, Read, Update, Delete)

Data Flow Between Components:

- Input Data: User prompts, login credentials
- Output Data: AI-generated responses, citations, error messages
- Stored Data: User profiles, saved chats
- Processed Data: Prompt validation, response formatting, citation generation

Data Sharing:

- Data is exchanged between frontend and backend via **RESTful APIs**
- Backend communicates with AI **API** through structured **JSON** requests/responses
- Database interactions are handled through secure queries
- No global shared memory is used; all communication is handled through **APIs**

3.4 Communications Interfaces

A.M.I. will exclusive interact with interact with the user through a website where the following protocols apply:

- **For Client-Server communication**
 - **HTTP**
 - **HTTPS**
- **For Secure data transmission**
 - **HTTP**
 - **HTTPS**

4. System Features

4.1 AI Chat Interaction

4.1.1 Description and Priority

This feature enables users to interact with the **A.M.I.** system through a text-based chatbot interface. Users can submit prompts related to personalized medicine and receive AI-generated responses enhanced with source transparency and citations.

Priority: **High**

4.1.2 Stimulus/Response Sequences

ID	Requirement
RS-01	User enters a prompt in the chatbot input field
RS-02	User submits the prompt
RS-03	System validates the input for readability and scope
RS-04	System displays a “thinking” state through text indicator
RS-05	System sends the prompt to the AI model
RS-06	AI model processes the request and generates a response
RS-07	System retrieves supporting sources
RS-08	System formats the response and citations (to generate references)

RS-09	System displays the response and references to the user
RS-10	System updates the avatar state based on response context

4.1.3 Functional Requirements

ID	Requirement
FRQ-01	The system MUST ensure that user can input a prompt
FRQ-02	The system MUST validate the user input is in a text readable form and within A.M.I. 's scope
FRQ-03	The system MUST display the state (idle, thinking, speaking) of the A.M.I. AI and avatar model
FRQ-04	The system MUST process the prompt and generate response
FRQ-05	The system MUST add disclaimer to response and create a reference of sources used

4.2 User Account Management

4.2.1 Description and Priority

This feature allows users to register, log in, and manage their accounts securely. It enables persistent access to saved chats and personalized interaction history.

Priority: **High**

4.2.2 Stimulus/Response Sequences

ID	Requirement
RS-11	User selects register or login option
RS-12	User enters required credentials (username, password)
RS-13	System validates input fields
RS-14	System securely stores or verifies credentials using encryption/hashing System creates or authenticates user session
RS-15	System redirects user to the main chat interface
RS-16	System provides logout functionality to end session

4.2.3 Functional Requirements

ID	Requirement
FRQ-06	The system MUST allow users to register an account by providing required credentials (e.g., username and password).
FRQ-07	The system MUST validate all user input fields during registration and login (e.g., non-empty, correct format).

FRQ-08	The system MUST securely store user credentials using encryption or hashing mechanisms.
FRQ-09	The system MUST authenticate users by verifying provided credentials during login.
FRQ-10	The system MUST create and manage a secure user session upon successful login. The system MUST allow authenticated users to log out and terminate their session securely.
FRQ-11	The system MUST restrict access to account-specific features (e.g., saved chats) to authenticated users only.
FRQ-12	The system MUST provide appropriate error messages when login or registration fails (e.g., incorrect credentials).
FRQ-13	The system MUST ensure that each user account is uniquely identifiable.
FRQ-14	The system MUST maintain session persistence for the duration of user activity, within defined security limits.

4.3 Chat History Management

4.3.1 Description and Priority

This feature allows logged-in users to save, retrieve, and manage previous chat sessions for continued interaction and review.

Priority: **High**

4.3.2 Stimulus/Response Sequences

ID	Requirement
RS-17	User logs into the system
RS-18	System retrieves stored chat sessions from the database
RS-19	System displays list of previous chats in sidebar
RS-20	User selects a chat
RS-21	System loads and displays messages for that chat
RS-22	User continues conversation or creates a new chat
RS-23	System saves new messages to the database

4.3.3 Functional Requirements

ID	Requirement
FRQ-15	The system MUST allow logged-in users to save chat conversations automatically or upon user action.
FRQ-16	The system MUST store chat messages and session data in a database associated with the correct user account.
FRQ-17	The system MUST retrieve stored chat sessions when a user logs into the system.

FRQ-18	The system MUST display a list of previous chat sessions in a clear and organized format (e.g., sidebar).
FRQ-19	The system MUST allow users to select a previous chat session.
FRQ-20	The system MUST load and display the full message history of the selected chat session.
FRQ-21	The system MUST allow users to continue an existing conversation within a selected chat session.
FRQ-22	The system MUST allow users to create a new chat session.
FRQ-23	The system MUST save new messages to the correct chat session in real time or near real time.

4.4 Source Transparency and Citation Display

4.4.1 Description and Priority

This feature ensures that AI-generated responses include supporting references to improve transparency and reduce the “black box” nature of AI.

Priority: **High**

4.4.2 Stimulus/Response Sequences

ID	Requirement
RS-24	User submits a prompt
RS-25	System determines whether external sources are needed
RS-26	System retrieves relevant sources from database
RS-27	System formats citations
RS-28	System attaches references to the response
RS-29	System displays references beneath the AI response

4.4.3 Functional Requirements

ID	Requirement
FRQ-24	The system MUST determine whether a response requires supporting references based on the user's prompt.
FRQ-25	The system MUST retrieve relevant sources used to support the generated response.
FRQ-26	The system MUST ensure that all retrieved sources are appropriate and related to the user's query.
FRQ-27	The system MUST format all references according to APA 7 citation guidelines.
FRQ-28	The system MUST attach the formatted references to the corresponding AI-generated response.
FRQ-29	The system MUST display references clearly beneath or alongside the AI response.

FRQ-30	The system MUST visually separate the main response content from the reference section for readability.
FRQ-31	The system MUST ensure consistency between the response content and the cited sources.
FRQ-32	The system MUST handle cases where sources are unavailable by notifying the user appropriately.
FRQ-33	The system MUST ensure that references are presented in a readable and accessible format for all users.

4.5 Prompt Validation and Safety Handling

4.5.1 Description and Priority

This feature ensures that user inputs are within the defined scope of the system and prevents unsafe or inappropriate use, including medical diagnosis requests.

Priority: **High**

4.5.2 Stimulus/Response Sequences

ID	Requirement
RS-30	User submits a prompt
RS-31	System analyzes prompt for scope and safety: • If prompt is valid: System proceeds with normal AI processing • If prompt is out-of-scope or unsafe: System generates a safe fallback response
RS-32	System displays disclaimer or guidance message
RS-33	System prevents diagnostic or prescriptive outputs

4.5.3 Functional Requirements

ID	Requirement
FRQ-34	The system MUST analyze all user prompts to determine whether they fall within the supported scope of A.M.I.
FRQ-35	The system MUST identify prompts that are unsafe, inappropriate, or requesting medical diagnosis or treatment.
FRQ-36	The system MUST allow valid prompts to proceed to normal AI processing.
FRQ-37	The system MUST block or redirect prompts that are out-of-scope or unsafe.

FRQ-38	The system MUST generate a safe fallback response when a prompt is invalid, unsafe, or outside scope.
FRQ-39	The system MUST provide a clear message explaining why the prompt cannot be processed when applicable.
FRQ-40	The system MUST display an appropriate disclaimer or guidance message for flagged prompts.
FRQ-41	The system MUST prevent the generation of responses that provide medical diagnosis or prescriptive treatment advice.
FRQ-42	The system MUST ensure that all responses remain within the educational and informational scope of the system.

4.6 Avatar Response and Visualization

4.6.1 Description and Priority

This feature provides visual feedback through a **Live2D** avatar that reacts to system states such as thinking, responding, or detecting urgency, and enhancing user engagement.

Priority: **Medium**

4.6.2 Stimulus/Response Sequences

ID	Requirement
RS-34	User submits a prompt System enters “thinking” state
RS-35	Avatar displays thinking animation (e.g., question mark effect)
RS-36	System processes response
RS-37	Avatar updates expression based on response tone or severity
RS-38	Avatar resets to idle state after interaction

4.6.3 Functional Requirements

ID	Requirement
FRQ-43	The system MUST display a Live2D avatar within the user interface during interactions.
FRQ-44	The system MUST transition the avatar to a “thinking” state when a user submits a prompt.
FRQ-45	The system MUST display a visual animation (e.g., question mark or similar effect) to indicate the thinking state.
FRQ-46	The system MUST update the avatar’s expression based on the response tone or detected level of concern (e.g., neutral, concerned, urgent).
FRQ-47	The system MUST synchronize avatar state changes with system processing states (thinking, responding, completed).
FRQ-48	The system MUST return the avatar to an idle state after the response has been delivered.

FRQ-49	The system MUST ensure that avatar animations do not interfere with the readability or usability of the interface.
FRQ-50	The system MUST handle cases where avatar rendering fails by maintaining core chatbot functionality without visual disruption.
FRQ-51	The system MUST maintain consistent avatar behavior across different user sessions and interactions.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

The **A.M.I.** system **SHALL** meet the following performance requirements to ensure efficient and reliable operation:

- **Response Time**
 - The system **SHOULD** generate and display AI responses within 60 seconds under normal operating conditions
 - Response time **MAY** vary depending on model size, system hardware, and network latency
 - The system **SHALL** provide visual feedback (e.g., “Generating response...” status and avatar thinking animation) while processing user input
- **System Throughput**
 - The system **SHALL** support continuous user interaction without requiring page refreshes
 - The system **SHALL** handle multiple consecutive prompts within a session without degradation in performance
- **Hardware Performance Constraints**
 - The system **DEPENDS** on sufficient hardware resources, including **GPU** capabilities, system **RAM**, and storage
 - Optimal performance is achieved on systems with a dedicated **GPU** (ex. **NVIDIA GPU** with **CUDA** support)
- **Systems without adequate GPU resources MAY experience**
 - Increased response latency
 - Reduced model performance
 - Reliance on **CPU**-based processing
- Large AI models **MAY** require significant **VRAM** and system memory, and insufficient memory **MAY** result in execution errors or failure to load the model

5.2 Safety Requirements

5.2.1 Medical Disclaimer and User Awareness

- The system **SHALL** make clear that it cannot officially diagnose any medical conditions
- The system **SHALL** make clear that it is not a medical professional and **WILL** recommend the user seek professional assistance if necessary

5.2.2 Restricted Medical Functionality

- The system **SHALL** not provide medical diagnosis, treatment, or prescriptions to the user under any circumstances
- The system **SHALL** recommend the user to seek professional assistance immediately if necessary

5.2.3 Handling harmful or High-risk prompts

- The system **SHALL** identify and safely respond with any prompts involving:
 - Suicide
 - Medical emergencies
 - Dangerous or unsafe health practices
- The system **SHALL** not generate content that could directly or indirectly cause harm to the user

5.3 Security Requirements

5.3.1 Data Safety and Privacy

- As per the Health Insurance Portability and Accountability Act of 1996 (**HIPAA**) and the AI Accountability and Personal Data Protection Act of 2025, **A.M.I. SHALL** not include any personally identifiable information related to users. This also includes any patient data that **MAY** be present in the sources used to train the model.

5.3.2 User Authentication

- The system **SHALL** allow guests limited access to the product without the need for authentication through login
- The system **SHALL** require user authentication (username/password) for access to account-specific features (such as saving and viewing chat history)
- The system **SHALL** require the user to abide by the secure password policies (minimum length and complexity requirements)

5.3.3 Data Storage

- The system **SHALL** store the username, password, and all chats with **A.M.I.** in a NoSQL document database in a secure manner
- The system **SHALL** store a hashed versions of the user's password in the database for security purposes

5.4 Software Quality Attributes

5.4.1 Software Usability

- The system **SHALL** provide a clear and intuitive web interface that allows users to submit prompts and read responses without prior training
- The system **SHALL** generate responses in a readable format for the user (English)

- The system **SHALL** ensure that users can start interacting with **A.M.I.** within two minutes of opening the home page
- The system **SHALL** include disclaimers and source messages for clarity and safety

5.4.2 Software Reliability

- The system **MUST** achieve at least 90% successful response generation rate under standard conditions
- The system **SHALL** handle all failures gracefully by displaying errors messages without crashing

5.4.3 Software Availability

- The system **SHALL** be available if the user is executing the program

5.4.4 Software Performance

- The system **MUST** generate AI responses in 60 seconds if the user is using a **NVIDIA** graphics card

5.4.5 Software Maintainability

- The system **SHALL** be structured in a modular design to allow for easier debugging and readability.
- The system **SHALL** include clear documentation for developers
- The system **SHALL** allow updates to the AI model, citation logic, and prompt constraints without needing a major system redesign

5.4.6 Software Transparency

- The system **SHALL** provide a clear and consistent citation display for all generated responses
- The system **SHALL** ensure that users can trace back responses to their corresponding sources
- The system **MUST** prioritize short, concise responses over complex and opaque responses

5.5 Business Rules

5.5.1 User Access Rules

- Guest users **WILL** be able interact with **A.M.I.** but cannot save chat history
- Registered users **MUST** be able to save, view, and manage their chat history
- Users **MUST** be able to create an account to access persistent features

5.5.2 Content Scope Rules

- The system **SHALL** respond properly to prompts related to:
 - Personalized medicine
 - General health education
- The system **SHALL** not accept prompts outside of scope rules

5.5.3 Data Handling Rules

- The system **MUST** not prompt the user for any personal medical data.
- User data **MUST** only be used for system functionality
- Users **MAY** request the deletion of personal data

5.5.4 Source Transparency Rules

- All generated responses from **A.M.I.** **MUST** include citations when necessary
- All citations **MUST** follow **APA 7** format
- The system **MUST** not present unsupported claims without references

6. Other Requirements

All the requirements for the **A.M.I.** software product have been covered in the SRS.

Appendix A: Glossary

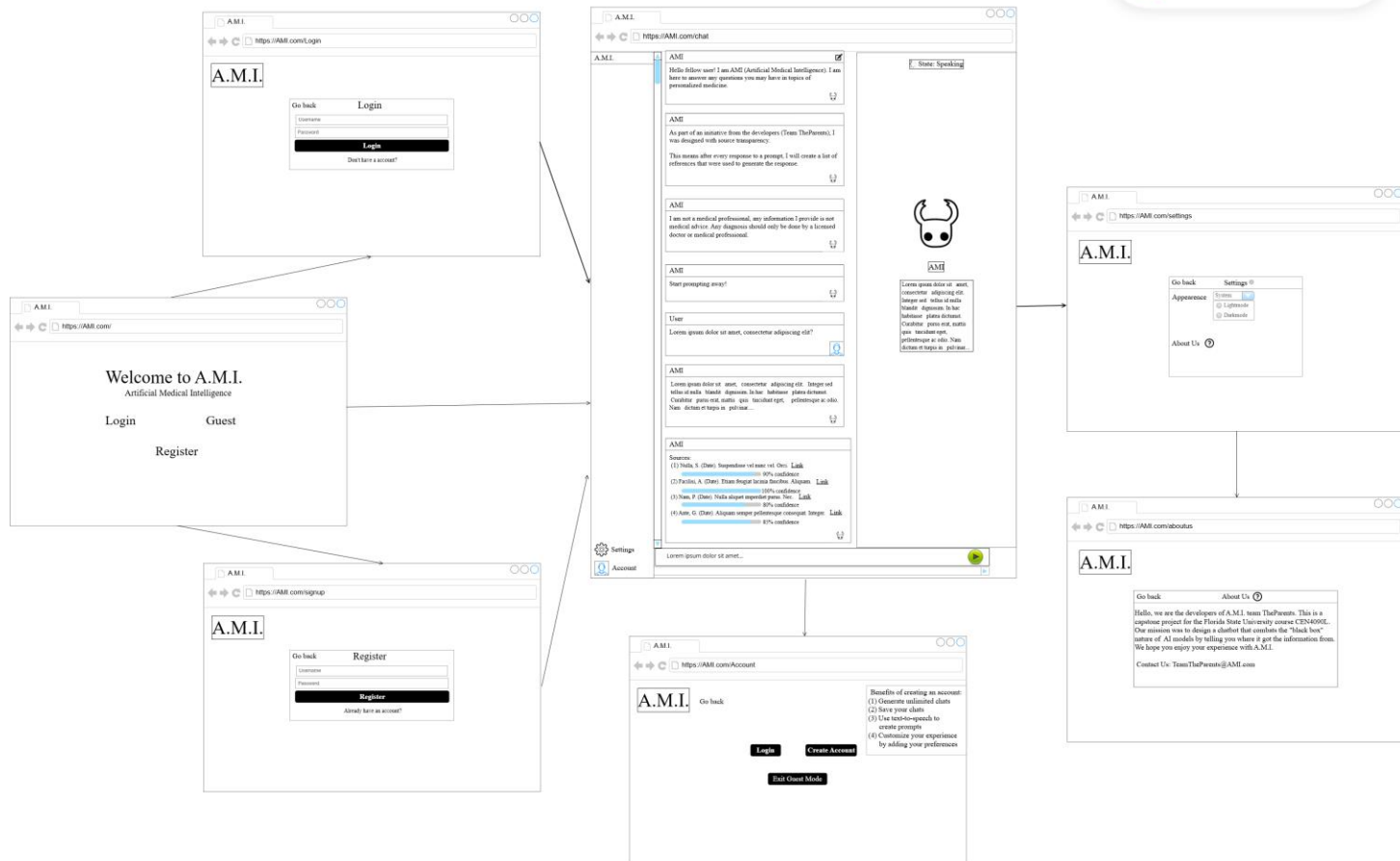
Term	Definition
A.M.I.	Artificial Medical Intelligence, AI-powered assistance developed in this system
APA 7	American Psychological Association 7 th edition citation format used for references
API	Application Programming Interface: a set of rules for communication between software components
BioPython	A python library used for biological and medical data processing, including access to PubMed
CDC	Center for Disease Control and Prevention; U.S. Public Health agency providing medical information
CDN	Content Delivery Network: A distributed system used to deliver web resources efficiently
CRUD	Create, Read, Update, Delete; basic operations for managing data in a system
CSS	Cascading Style Sheets; used to style and design web pages

Disclosure Policy	A statement informing users that A.M.I is not a medical professional and provides informational content only
Entrez API	An API by the National Library of Medicine for accessing PubMed and other databases
Flask	A lightweight Python framework used to build the backend of A.M.I
GPU	Graphics Processing Unit; hardware used to accelerate computation, especially for AI models
HIPAA	Health Insurance Portability and Accountability Act; A U.S. law that establishes standards for protecting sensitive patient health information and data privacy and security
HTML	HyperText MarkUp Language; used to structure web page content
HTTP	HyperText Transfer Protocol; used for communication between web clients and servers
HTTPS	Secure version of HTTP that encrypts data transmission
HuggingFace	A platform providing AI models, datasets, and tools for machine learning applications
JavaScript	A programming language used for client-side interactivity in web applications
JSON	JavaScript Object Notation; a lightweight data format used for data exchange
Live2D	A technology used to animate 2D characters, used for the A.M.I. avatar
LLaMa	Large Language Model Meta AI; a family of language models developed by Meta
LLM	Large Language Model; an AI model trained to understand and generate human-like text
MongoDB	A NoSQL database used to store user accounts and chat history
NIH	National Institutes of Health; a U.S. medical research agency
Node	A runtime environment (Node.js) used for executing JavaScript outside the browser
NoSQL	A type of database that stores data in non-relational formats
NVIDIA	A company that produces GPUs commonly used for AI and machine learning tasks
PEFT	Parameter-Efficient Fine-Tuning; a method for fine-tuning large models efficiently
Personalized Medicine	Medical approach that tailors treatment and information to individual characteristics
PixiJS	A JavaScript library used for rendering graphics, used with Live2D in the frontend

PubMed	A database of biomedical literature maintained by the National Library of Medicine
Python	A programming language used for backend development and AI processing
PyTorch	A machine learning framework used to build and run AI models
RAM	Random Access Memory; temporary system memory used during processing
RESTful API	An API that follows REST principles for communication over HTTP
Source Transparency	The practice of providing sources or references used to generate AI responses
Transformer	A type of neural network architecture used in modern language models
UI	User Interface; the visual components users interact with
UI/UX	User Interface and User Experience; design and usability aspects of the system
Unsloth	A framework used to optimize and fine-tune LLaMA-based models efficiently
VPM	Virtual Patient Model; a simulated model used for medical interaction or testing (if used in your context)
VRAM	Video Random Access Memory; memory used by GPUs for processing large models

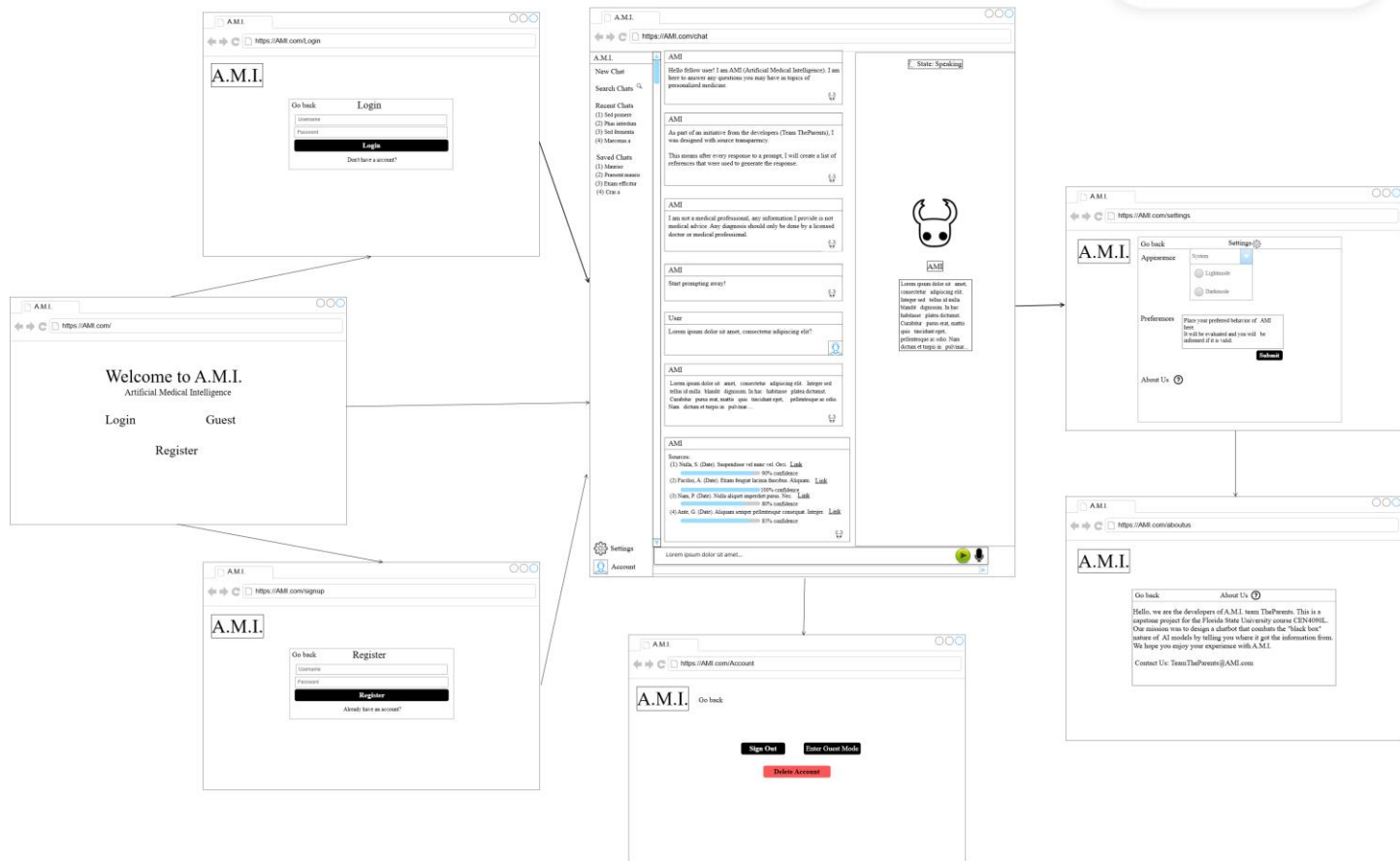
Appendix B: Analysis Models

AMI (Guest Mode)



UI/UX Wireframe (Logged-In mode):

AMI (Account Login)



Appendix C: To Be Determined List

There are no more references that need to be tracked for closure.